

Sujan Parajuli

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Professional Summary:

Dedicated Certified (GISP 162370) Geospatial Data Professional with proven expertise in remote sensing, geospatial data management, and GIS application development for federal and environmental programs. Skilled in designing scalable data pipelines, implementing AI/ML models for spatial-temporal analysis, and deploying production-ready GIS solutions in both enterprise and cloud environments. Proficient in leveraging R, SQL, and Excel for advanced data analysis and visualization (R Studio, Python, and ArcGIS Dashboards). Experienced working with multi-agency teams at USGS, NOAA NMFS, USFWS, and AZDEQ to deliver geospatial products that support conservation, natural resource management, and data-driven decision-making.

Work Experience:

GIS Consultant | Arizona Department of Environmental Quality (AZDEQ) | Mar 2025 – May 2026

- Developed deployable web mapping applications with ArcGIS Enterprise and Python to support environmental data analysis and business decisions.
- Created and maintained spatial and non-spatial datasets, ensuring data accuracy, completeness, and consistency within centralized data libraries.
- Designed geospatial visualizations and dynamic maps to communicate environmental insights to technical and non-technical audiences.
- Use Excel, and R for data analysis, and create data visualizations in R Studio for modeling and mapping community assessment for Arizona to support environmental reporting.
- Designed and deployed a Survey123-based data collection tool for recruiting and managing agency volunteers.
- Support data management tasks related to PFAS sampling, ensuring data quality and accessibility for environmental monitoring.
- Conducted spatial analysis workflows and developed metadata, standard operating procedures, and process documentation.
- Collaborated with cross-functional teams to align geospatial solutions with business goals, providing GIS training and user support.

Senior Geospatial Analyst | Ocean Associates Inc., contractor to NOAA | October 2024 - March 2025

- Assisted with the design, creation, and management of geospatial databases to support the analysis of environmental and marine ecosystems.
- Integrated complex data, including GPS records, shapefiles, field measurements, and geophysical data, into centralized GIS platforms for seamless access and analysis.
- Conducted advanced 3D spatial analysis and geospatial modeling to measure and predict the impacts of environmental factors, such as human activity and climate change, on protected species.
- Translated raw environmental data into actionable insights using GIS tools, identifying spatial patterns to improve conservation strategies.
- Created detailed maps, both hard and soft copies, for reports that showcased species density, anthropogenic threats, and oceanographic features. These maps were instrumental in conveying complex environmental data to stakeholders.
- Built analytical workflows in R, and SQL in identifying trends, anomalies, and priority commodities at risk.
- Produced technical reports and visual dashboards for both internal NOAA use and public

GIS Data Analyst/Developer | Astor and Sanders, contractor to USFWS | May 2024 - Present

- Offered GIS technical assistance, expertise, and recommendations adhering to federal policies, procedures, regulations, and laws regarding technology.
- Collaborated directly with conservation biologists to ensure consistent interpretation by providing mapping support and curating spatial data across a species' range.
- Utilized R, Python, and Shiny to develop and implement data processing pipelines and interactive applications for species range mapping.
- Provided expert support in developing and refining statistical and machine learning algorithms being deployed in data products/applications/services.
- Coordinated with various offices to determine the appropriate scale for mapping shared species.
- Conducted quality assurance and quality control on range projects in accordance with established SOPs and refined SOPs based on feedback from project participants.

Remote Sensing Data Scientist II | KBR Inc., contractor to USGS EROS | Oct 2018 - May 2024

- Applied statistical analysis and machine learning techniques to geospatial data, developing and refining algorithms for data products/applications/services.
- Utilized Python and R for data processing, analysis, and modeling, including the implementation of machine learning frameworks such as TensorFlow and scikit-learn.
- Developed and optimized code for processing large environmental datasets, both structured and unstructured.
- Developed and maintained GIS applications using Python for geospatial data processing, and utilized ArcGIS and QGIS for spatial analysis and data visualization in Tableau and Power BI.

- Implemented Python scripts for automated and semi-automated interpretation of remotely sensed data, including lidar data for vegetation and fuel characteristics assessment.
- Led the LiDAR Wildland Fuels project, employing pdal, Visual Studio, and Git for version control, and utilizing SQL for efficient data retrieval and manipulation.
- Utilized high-performance computing resources to process large datasets for image classification, developing scalable image processing pipelines.
- Applied advanced modeling techniques, including regression, supervised learning, unsupervised learning, probabilistic inference, and natural language modeling for accurate and efficient data analysis.
- Managed and processed data from various satellite sensors and images, including Landsat, MODIS, and Sentinel, ensuring accurate and timely data delivery.
- Coordinated and engaged with other federal agencies, states, nongovernmental organizations, and interested parties to share data informing species range delineations.
- Knowledge of heritage IDL, contributing to legacy data processing and analysis tasks.

Teaching Assistant | South Dakota State University, Department of Geography | August 2015 – May 2018

- Facilitated and taught over 500 students, working closely to help them excel in Data Creation and Integration, Physical Geography, and Weather and Systems labs.
- Provided technical support for ESRI software to students and delivered pre-lab talks, demos, and training on using ArcGIS for problem-solving.
- Collaborated with research scientists, geographers, professors, and students across disciplines to identify methodological challenges in communicating geospatial tools.
- Demonstrated excellent analysis and research skills, written/verbal communication, and organizational skills.

GIS Intern | Mid-Dakota Vegetation Management, Miller, SD | May - August 2017

- Utilized ESRI Suites and Model Builder for GIS data research, analysis, and development.
- Applied Python for developing tools and tracking and reviewing GIS data.
- Created and maintained the data database using ArcInfo, Python Scripting, SQL, and R2V, reducing costs by 150%.
- Produced and prepared GIS-related maps, work plans, and reports using ArcMap, ArcCatalog, Model Builder, and PowerBI.
- Developed and maintained GIS databases, ensuring data integrity, and contributed to developing and maintaining GIS documentation and standard operating procedures.

Education:

M.S. in Geography (GIS Specialization) with Data Science Certificate

South Dakota State University, Brookings, SD

B.Sc. in Forestry

Tribhuvan University, Institute of Forestry, Pokhara, Nepal

Certification:

Certified GISP July 2025 - July 2028

Certification Number: 162370

Technical Skills:

Programming Languages: Python, R, JavaScript, HTML, CSS

GIS: ArcGIS Desktop, ArcGIS Online, ArcGIS Pro, QGIS

Data Visualization: Tableau, Power BI

Machine Learning: TensorFlow, scikit-learn, XGBoost

Web GIS Development: ArcGIS Server/SDE, WebGIS, GeoServer, PostGIS, ArcGIS API for Python/JavaScript

Scientific Data Processing: HDF, netCDF

Version Control: GitHub, GitLab

Database Management: PostgreSQL, MySQL, SQL Server, BigQuery, Excel

Platforms: Linux, Windows

Cloud Computing: AWS, Google Earth Engine, High-Performance Computing (HPC)

Agile Tools: Jira

IDEs: Visual Studio, Jupyter Notebook, PyCharm, Anaconda

Microsoft Office Suite: MS Visio, MS Word, MS PowerPoint, MS Excel, MS Access